SentinelIQ 04 — Data Strategy & Scope

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SentinelIQ cannot use real company security data, so v1 normalizes two sources into one event schema: public network flows and synthetic authentication logs.

Why two sources

A single public dataset would prove the ML layer but not the product, because it has no users, hosts or login activity to reason about. Two sources cover both needs.

- **Public network flows** give labeled attack and benign traffic, which is what a supervised model needs for honest metrics.
- **Synthetic authentication logs** give identities, failed logins and unusual access patterns, which let the RAG triage layer explain behaviour in terms a SOC analyst recognizes (for example credential attacks).
- **One schema** forces the ingestion and normalization design that real security products need.

Source A: public network flows

The plan is to use CICIDS2017, a public dataset from the Canadian Institute for Cybersecurity. It contains labeled network flows with roughly 80 numeric features per flow, covering benign traffic and attack families such as brute force, DoS and DDoS, port scans, web attacks and botnet activity (details to be confirmed against the dataset documentation before Phase 4).

- **Use:** training and evaluating the anomaly model in Phase 4.
- **Handling:** flows are converted into the unified schema; the raw feature vector is kept in a JSON column.
- **Size:** a sampled subset will be used so that the project runs on a laptop with Docker.

Source B: synthetic authentication logs

A small generator script will produce login events for a fictional organization with a fixed random seed, so results are reproducible. It must simulate both normal behaviour and a short list of attack patterns that map to ATT&CK techniques.

- **Normal behaviour:** office-hours logins, a few devices per user, occasional typos in passwords.
- **Brute force:** many failed logins against one account from one address.
- **Password spraying:** a few attempts each across many accounts from one address.
- **Unusual access:** a successful login from a new country or at an odd hour, followed by sensitive actions.

Every generated event carries its true label and the pattern that produced it, which makes explanation accuracy measurable.

Unified event schema

This is the first draft of the schema; Document 08 (Data Model & API Specification) will finalize types and indexes.

Field	Type	Meaning
event_id	UUID	Unique identifier assigned on ingestion
dedup_key	text	Hash of the source event, used to reject duplicates
source	enum	`network_flow` or `auth_log`
occurred_at	timestamp	When the event happened at the source
ingested_at	timestamp	When SentinelIQ received it
src_ip, dst_ip	text	Source and destination addresses where present
src_port, dst_port	integer	Ports where present
protocol	text	Network protocol where present
username	text, nullable	Account involved, for authentication events
event_type	text	Normalized type, such as `flow` or `login_failed`
features	JSON	Source-specific numeric features used by the model
raw	JSON	The original event, kept untouched
label	text, nullable	Ground-truth label, used only for evaluation

Labels and evaluation split

Honest metrics depend on keeping test data away from training.

- **Split before anything else:** the test set is set aside first and is not used for tuning thresholds or prompts.
- **Keep related flows together:** where the dataset allows, the split separates traffic by capture day or session so that near-duplicate flows do not appear on both sides.
- **Class imbalance:** report per-class precision and recall, not accuracy alone, because benign traffic dominates.
- **Explanation set:** a small hand-labeled set of events, each paired with the correct ATT&CK technique, is used to measure whether explanations cite the right technique. Its size will be fixed in Document 09.
- **Labels in production paths:** the `label` field is read only by evaluation code, never by the scoring or triage services.

Limitations to state openly

- Public benchmark traffic is older and cleaner than real enterprise traffic, so scores on it are not a promise about production.
- Synthetic logs reflect the patterns the generator author imagined, so they test the pipeline and the explanations, not real attacker creativity.
- The labeled attack classes are a small subset of what real attackers do.
- Because synthetic and public data come from different worlds, v1 evaluates each source separately and does not claim a single combined accuracy figure.
- Dataset licensing and citation requirements will be checked and recorded in the README before release.